

QUANTIZATION OF GRAVITATION WITH WEYL FERMIONS

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It is shown that quantization of gravitation consistent with the presence of Weyl fermions is possible, in spite of the existence of Lorentz anomalies: the group of local Lorentz transformations is quantized becoming a physical field and the anomaly is absorbed.

1. Introduction

It has become clear after the beautiful paper of Polyakov on the quantization of strings,¹ that in theories possessing a gauge invariance, the associated degree of freedom—decoupled at the classical level—may appear as physical after quantization.

In the case of Yang-Mills theories with Weyl fermions, this phenomenon is responsible for the nonanomalous behavior of the quantum theories, as shown independently—in the path integral approach—by Kulikov,² Harada and Tsutsui³ and Babelon, Viallet and one of the present authors.⁴ Indeed, the proper treatment of the gauge degrees of freedom uncovers the presence of the Wess-Zumino action and the group of gauge transformations gets quantized, becoming a *physical field*.⁴ Remarkably, the resulting generating functional is independent of the choice of gauge section and hence the resulting theory is nonanomalous.

When Weyl fermions propagate in an external gravitational field in a space-time of dimension $2(\bmod 4)$ gravitational anomalies exist, as shown by Alvarez-Gaumé and Witten.⁵ These anomalies are associated with the violation of general coordinate invariance (Einstein invariance) and local Lorentz invariance. Since Einstein anomalies can always be transformed into Lorentz anomalies,⁶ all gravitational anomalies can be seen as violations of local Lorentz invariance and this will be the view we shall take in the present investigation.

In this work we consider Weyl fermions coupled to gravitation and investigate, following the proposal presented in Ref. 4, the resulting quantum theory. We start from the generating functional Z written as a functional integral over the whole field space rather than as an integral over the fields in a definite gauge (orbit space). When only Dirac fermions are present there is no harm in looking at Z as an integral over the orbit space since the integration over the group of transformations (associated with Einstein and Lorentz symmetries) can be factorized by means of the usual Faddeev-Popov procedure.^{7,8,9} On the contrary, when Weyl fermions are present—and because of their ability to create gravitational noninvariant interactions—this is not the case

anymore. Our main result is that local Lorentz transformations acquire the status of a physical field which enters in Z in a nontrivial effective action and the resulting quantum theory is nonanomalous in the sense that Z does not depend on the condition fixing local Lorentz frames.

We present in Sec. 2 the general treatment and then discuss in Sec. 3, a simple two-dimensional example.

2. The General Treatment

We start from the path integral version of the generating functional Z in a n -dimensional Euclidean space, following Fujikawa's choice for the gravitational measure¹⁰ (see also Refs. 11 and 12). Z reads

$$Z = \int \mathcal{D}\mu(g) \mathcal{D}[g^{1/4}\bar{\psi}] \mathcal{D}[g^{1/4}\psi] \exp\{-S[\bar{\psi}, \psi, h]\} \quad (2.1)$$

where¹⁰

$$\mathcal{D}\mu(g) = \prod_{v,a} \mathcal{D}[g^k h_v^a], \quad k = \frac{n-2}{4}, \quad (2.2)$$

with $h_v^a(x)$ the vierbein fields,

$$h_v^a(x) h_{\mu a}(x) = g_{\nu\mu}(x) \quad (2.3)$$

and

$$g = \det g_{\nu\mu}(x) \geq 0 \quad (2.4)$$

in the Euclidean metric we are using.

Following Ref. 10, we are considering $\mathcal{D}[g^{1/4}\psi] = \mathcal{D}\tilde{\psi}$ and $\mathcal{D}[g^{1/4}\bar{\psi}] = \mathcal{D}\tilde{\bar{\psi}}$ as the fermions measures in the presence of gravitational fields. The action S is

$$S = S_G + \int d^n x \det h_a^\mu \frac{i}{2} \bar{\psi} \gamma^a h_a^\mu \vec{D}_\mu \psi = S_G + \int d^n x \bar{\psi} D(h) \psi, \quad (2.5)$$

with

$$\vec{D}_\mu = \left(\vec{\partial}_\mu - \frac{i}{4} \sigma_{ab} \omega_\mu^{ab} \right) \frac{(1 - \gamma_5)}{2} \quad (2.6)$$

and the Lorentz connection ω_μ given by

$$\omega_\mu^{ab} = h^{va} \nabla_\mu h_v^b \quad (2.7)$$

with

$$\nabla_\mu v^\nu = (\partial_\mu \delta_\rho^\nu - \Gamma_{\mu\rho}^\nu) v^\rho. \quad (2.8)$$

The explicit form of the gravitational action S_G will not be necessary for the present analysis.

The action S is invariant both under general coordinate transformations

$$x_\mu \rightarrow x_\mu + \xi_\mu(x) \quad (2.9)$$

as well as under local Lorentz transformations

$$x_\mu \rightarrow \Lambda_\mu^\nu(x) x_\nu. \quad (2.10)$$

One then has to carefully treat the functional integration in (2.1) in order to get a sensible result. When Dirac fermions are present this amounts to extracting the gauge orbit volume element so that Z can be viewed as an integral over orbit space. The problems arising due to the peculiarities of gravitation when treated as a gauge theory⁹ can be overcome either by exploiting BRS supersymmetry, which is one of the consequences of the Faddeev-Popov effective action,^{13,14,15} or by a careful treatment of the “resolution of the unity”^{10,11,12} if closely following the Faddeev-Popov approach.⁷ We shall take this latter way of quantizing the theory.

Now, if Weyl fermions are present the procedure has to be revised since, as it happens with Yang-Mills theory, the presence of a Fujikawa Jacobian¹⁵ prevents the factorization of the gauge orbit volume. We shall precisely discuss this fact in detail in what follows.

As it has been shown by Bardeen and Zumino⁶ it is always possible to define the generating functional so that general coordinate invariance is respected. Then, concerning this symmetry, the standard treatment works even if Weyl fermions are present and hence we shall be concerned with the local Lorentz invariance only.

In the usual Popov-Faddeev approach, adapted to the gravitational case by Fujikawa and Yasuda,¹¹ one uses a “resolution of unity” in order to factorize the invariant gauge orbit volume element. In the present case we write

$$1 = \int \delta(F_2[g_{\mu\nu}^\sigma]) \delta(F_1[h_a^{\mu;v,\lambda}]) \Delta_{FP}(g_{\mu\nu}; h_\mu^a) \mathcal{D}v_\mu \mathcal{D}\lambda, \quad (2.11)$$

where v_μ is a vector which describes the gauge orbit corresponding to general coordinate transformations and $\mathcal{D}\lambda$ represents the integration over the local Lorentz transformations

$$h_a^{\mu;\lambda} = h_a^\mu - \lambda_a^b h_b^\mu, \quad (2.12)$$

Δ_{FP} is the Faddeev-Popov determinant associated with the gauge conditions for Lorentz and Einstein invariances

$$F_1[h_\mu^a] = 0, \quad (2.13)$$

$$F_2[g_{\mu\nu}] = 0. \quad (2.14)$$

The invariant volume elements to be used are:¹¹

$$\mathcal{D}(g^{(n+2)/4n} v_\mu), \quad (2.15)$$

$$\mathcal{D}(\sqrt{g}\lambda), \quad (2.16)$$

when introducing (2.11) into (2.1). As we stated above, the invariant gauge orbit volume element $\mathcal{D}(g^{(n+2)/4n} v_\mu)$ can be factorized using the invariance of *all* measures (as well as that of Δ_{FP} and S) under Einstein transformations. However, when trying to factorize the λ -integration one has to write

$$\mathcal{D}(g^{1/4} \bar{\psi}) \mathcal{D}(g^{1/4} \psi) = J(\lambda, h) \mathcal{D}(g^{1/4} \bar{\psi}^\lambda) \mathcal{D}(g^{1/4} \psi^\lambda), \quad (2.17)$$

where $\bar{\psi}^\lambda, \psi^\lambda$ are the Lorentz transforms of $\bar{\psi}, \psi$. This is due to the fact that it is not possible to define a measure both Einstein and Lorentz invariant for Weyl fermions. J is the Fujikawa Jacobian which depends on h_μ^a through the regularization procedure (see below).

After re-labeling variables the result for Z is

$$Z = \int \mathcal{D}(g^k h_\mu^a) \mathcal{D}(g^{1/4} \bar{\psi}) \mathcal{D}(g^{1/4} \psi) \mathcal{D}(\sqrt{g}\lambda) \delta(F_1) \delta(F_2) \Delta_{FP} J(\lambda, h^{\lambda^{-1}}) \exp(-S). \quad (2.18)$$

The presence of the Jacobian prevents in general, the factorization of the Lorentz gauge orbit volume element and hence, in those cases in which $J \neq 1$ (which corresponds to space-time dimensions $n = 2 + 4m, m \in \mathbb{Z}$) the λ -field acquires the status of a physical field.

From (2.17) we easily see that

$$J(\lambda, h) = \frac{\det D(h)}{\det D(h^\lambda)}, \quad (2.19)$$

where some regularization is understood in the right-hand side of Eq. (2.19). Using this relation, it can immediately be seen that Z does not depend on the gauge choices (2.13) and (2.14) and hence the resulting gravitational theory defined from it is non-anomalous. Indeed, define an effective action by integrating over fermions and the λ -field

$$\exp\{-S_{\text{eff}}[h]\} = \int \mathcal{D}(g^{1/4} \bar{\psi}) \mathcal{D}(g^{1/4} \psi) \mathcal{D}(g^{1/2} \lambda) J(\lambda, h^{\lambda^{-1}}) e^{-\int \bar{\psi} D(h) \psi d^n x}. \quad (2.20)$$

Under a local Lorentz transformation

$$h \rightarrow h^\sigma, \quad (2.21)$$

one gets

$$\exp\{-S_{\text{eff}}[h^\sigma]\} = \int \mathcal{D}(g^{1/4}\bar{\psi})\mathcal{D}(g^{1/4}\psi)\mathcal{D}(g^{1/2}\lambda)J(\lambda, h^{\sigma\lambda^{-1}})e^{-\int \bar{\psi}D(h)\psi d^n x}. \quad (2.22)$$

Now, using that $D(h)$ is a covariant derivative and performing the transformation

$$\begin{aligned} \bar{\psi} &\rightarrow \bar{\psi}^{\sigma^{-1}} \\ \psi &\rightarrow \psi^{\sigma^{-1}} \end{aligned} \quad (2.23)$$

one gets

$$\exp\{-S_{\text{eff}}[h^\sigma]\} = \int \mathcal{D}(g^{1/4}\bar{\psi})\mathcal{D}(g^{1/4}\psi)\mathcal{D}(g^{1/2}\lambda)J(\lambda, h^{\sigma\lambda^{-1}})J(\sigma^{-1}, h^\sigma)e^{-\int \bar{\psi}D\psi d^n x}. \quad (2.24)$$

However

$$J(\lambda, h^{\sigma\lambda^{-1}}) \cdot J(\sigma^{-1}, h^\sigma) = J(\sigma\lambda^{-1}, h), \quad (2.25)$$

and hence, using

$$\mathcal{D}(g^{1/2}\lambda) = \mathcal{D}(g^{1/2}\lambda\sigma), \quad (2.26)$$

the invariance of the effective action follows

$$S_{\text{eff}}(h^\sigma) = S_{\text{eff}}(h). \quad (2.27)$$

Note that the presence of the λ -integration was crucial in this proof. That this is so was first noted by Faddeev and Shatashvili,¹⁶ who defined a nonanomalous chiral gauge theory by the addition of a group-valued field with an adequate action to the original model. For gravitational theories the same fact was stressed by Keke Li¹⁷ in the study of two-dimensional gravity. It is important to note that in the present formulation the *ad hoc* addition of a group-valued field is not necessary: it naturally appears in the process of quantization (as it happens in the case of Yang-Mills theories^{2, 3, 4}).

We want to comment at this stage that it remains to be analyzed if the new degrees of freedom appearing in the correct treatment of the Lorentz symmetry give rise to a consistent unitary model. We shall see that this certainly happens in a two-dimensional

example, but the problem in a more realistic model is much more involved and is presently under investigation.

3. A Two-Dimensional Example

We consider in this section two-dimensional gravity coupled to chiral fermions as an application of the approach described above. Although ordinary Einstein gravity cannot be formulated in $n = 2$, a nontrivial action can be obtained by dimensional reduction from $n = 3$, as was first shown by Jackiw.¹⁸ We shall closely follow this approach and the analysis of anomalies given in Ref. 19 except that in our treatment Lorentz invariance at the gravitational level is assured without addition of any auxiliary field.

We write for completeness the dimensionally reduced two-dimensional gravitational action¹⁸

$$S_G = -\frac{1}{16\pi G} \int d^2x N \sqrt{g} (R + 2\Lambda), \quad (3.1)$$

where the scalar factor N is related to the original ($n = 3$) vierbein field, R is the curvature scalar and Λ a cosmological constant. The (reduced) two-dimensional fermion action is

$$S_F = \int d^2x \sqrt{g} \frac{i}{2} h_a^\mu \bar{\psi} \gamma^a \vec{\partial}_\mu \psi \quad (3.2)$$

with $\gamma_5 \psi = -\psi$, i.e. left-handed (for definiteness) fermions. The fermionic action does not explicitly depend on the spin connection. The equation of motion from (3.2) is

$$D(\omega)\psi = i\gamma^\mu (\partial_\mu + \frac{1}{2}\omega_{\mu,ab}\sigma^{ab})\psi = 0 \quad (3.3)$$

and the fermionic path integral is then defined as the determinant of $D(\omega)$.

In two-dimensional space-time, not only J given by (2.19) can be evaluated but each determinant separately can be computed exactly. This is due to the fact that two-dimensional Riemannian spaces are conformally flat and hence in a suitable local coordinate system we can repeat the standard steps leading to the exact two-dimensional fermion determinant.^{19,20} The answer, first given by Leutwyler²¹ is

$$\begin{aligned} \log \det D(\omega) = & \frac{-1}{192\pi} \int d^2x \left[\sqrt{g} R \frac{1}{\sqrt{g} \nabla^2} (\sqrt{g} R - \partial_\mu \sqrt{g} g^{\mu\nu} \omega_\nu) + \mu \sqrt{g} \right. \\ & \left. + \frac{a}{2} \sqrt{g} g^{\mu\nu} \omega_\mu \omega_\nu \right], \end{aligned} \quad (3.4)$$

where μ and a are free parameters related to the arbitrariness in the definition of the regularized determinant. Due to the presence of the second term in (3.4) no choice

of a makes the determinant invariant under Lorentz transformations (a plays a role analogous to the coefficient of the contact term A_μ^2 appearing in the chiral Schwinger model, which can be used, as shown by Jackiw and Rajaraman²² to render the model consistent).

From (3.4) and (2.19), $J(\lambda, h)$ can be computed. Concerning the effective action, note that according to (2.20)

$$\exp(-S_{\text{eff}}[h]) = \int \mathcal{D}(g^{1/2}\lambda) J(\lambda, h^{\lambda^{-1}}) \det D(h) = \int \mathcal{D}(g^{1/2}\lambda) \det D(h^{\lambda^{-1}}). \quad (3.5)$$

Hence, the effective action can be directly expressed in terms of (3.4) and of course it is gauge-invariant as can be easily seen from (3.4) and (3.5) by making

$$\omega_\mu \rightarrow \omega_\mu^\sigma + \partial_\mu \sigma. \quad (3.6)$$

If we write, instead of (3.5), the effective action in terms of fermions and the Jacobian, as in (2.20), but with its explicit expression (3.4), we have

$$\begin{aligned} \exp(-S_{\text{eff}}) = & \int \mathcal{D}(g^{1/4}\bar{\psi}) \mathcal{D}(g^{1/4}\psi) \exp \left\{ - \int \bar{\psi} D(h) \psi d^2x \right\} \mathcal{D}\theta \exp \left\{ \int d^2x \right. \\ & \times \left[\frac{a}{2} \partial_\mu \theta \partial_\nu \theta \sqrt{-g} g^{\mu\nu} + a \sqrt{-g} g^{\mu\nu} \omega_\mu \partial_\nu \theta + \frac{\beta}{192\pi} \theta \sqrt{-g} R \right] \left. \right\} \end{aligned} \quad (3.7)$$

where

$$\lambda = \begin{pmatrix} \cosh \frac{1}{2}\theta & \sinh \frac{1}{2}\theta \\ \sinh \frac{1}{2}\theta & \cosh \frac{1}{2}\theta \end{pmatrix}. \quad (3.8)$$

We then see, at first noted in Ref. 17, that θ describes a massless scalar particle for $a > 0$, while for $a < 0$ it is a ghost. In the first case the theory is consistent in the sense that the resulting effective Lagrangian is free of unphysical particles. One expects that in more realistic models a similar choice of the regularization parameters will make the new degrees of freedom represent physical fields.

Now, in two dimensions the effective action can be computed exactly too, since the functional integral (3.5) is Gaussian

$$S_{\text{eff}} = \frac{1}{192\pi} \int d^2x \left[\sqrt{g} R \frac{1}{\sqrt{g}} g^{\mu\nu} \omega_\mu \omega_\nu + (\partial_\mu \sqrt{g} g^{\mu\nu} \omega_\nu) \frac{1}{\sqrt{g} \nabla^2} (\partial_\rho \sqrt{g} g^{\rho\delta} \omega_\delta) \right] \quad (3.9)$$

In spite of being explicitly dependent on the connection, the nonlocal effective action (3.9) is invariant under the transformation (3.6).

In summary, under the light of the results of Refs. 2–4 and those presented here it is reasonable to state that the phenomenon we describe happens with any classical

symmetry: integration over the symmetry takes care of the anomaly and a group-valued field acquires physical status.

To have a complete comprehension of the resulting effective theory one has to evaluate J in the general case (with $m = 2 + 4m$) by integration of the Lorentz anomaly and then analyze the effective action with the g -field playing a physical role. Work on this subject will be presented elsewhere.

It is interesting to note that the present treatment can be extended to SUSY Yang-Mills theories coupled to supergravity, the low energy limit of certain string models. The cancellation of anomalies *à la* Green-Schwarz²³ for certain models will correspond, in the present approach, to the mutual cancellation of the Jacobians associated with each of the symmetries (gauge and gravitational). In this particular case the associated degrees of freedom factorize as in nonanomalous theories (where $J = 1$ separately).

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References

1. A. M. Polyakov, *Phys. Lett.* **103B** (1981) 207.
2. A. V. Kukilov, IHEP preprint 86-83 (unpublished).
3. K. Harada and I. Tsutsui, Tokyo Institute of Technology preprint. TIT-HEP-94, 1986 (unpublished).
4. O. Babelon, F. A. Schaposnik and C. M. Viallet, *Phys. Lett.* **177B** (1986) 385.
5. L. Alvarez-Gaumé and E. Witten, *Nucl. Phys.* **B324** (1984) 269.
6. W. A. Bardeen and B. Zumino, *Nucl. Phys.* **B244** (1984) 421.
7. L. D. Faddeev and V. N. Popov, *Phys. Lett.* **25B** (1967) 29.
8. B. de Witt, *Phys. Rev.* **160** (1967) 1113; **162** (1967) 1195; **162** (1967) 1195.
9. E. Fradkin and G. Vilkovisky, *Phys. Rev.* **D8** (1973) 4241 and references therein.
10. K. Fujikawa, *Nucl. Phys.* **B226** (1983) 437.
11. K. Fujikawa and O. Yasuda, *Nucl. Phys.* **B245** (1984) 436.
12. K. Fujikawa, M. Tomiya and O. Yasuda, *Z. Phys.* **C28** (1985) 289.
13. K. Stelle and P. West, *Nucl. Phys.* **B140** (1978) 285.
14. M. Fung, P. van Nieuwenhuizen and D. Jones, *Phys. Rev.* **D22** (1980) 2995.
15. K. Fujikawa, *Phys. Rev. Lett.* **42** (1979) 1195; **44** (1980) 1733, *Phys. Rev.* **D21** (1980) 1499; **D22** (1980) 1499 (E); **D23** (1981) 2262.
16. L. D. Faddeev and S. L. Shatashvili, *Phys. Lett.* **167B** (1986) 225.
17. Keke Li, *Phys. Rev.* **D34** (1986) 2292.
18. R. Jackiw, in *Quantum Theory of Gravity*, ed. S. Christensen (Hilger, Bristol, 1984) and *Nucl. Phys.* **B252** (1985) 343.
19. R. E. Gamboa Saraví, M. A. Muschetti, F. A. Schaposnik and J. E. Solomin, *Ann. Phys. (N.Y.)* **157** (1984) 360.
20. R. E. Gamboa, F. A. Schaposnik and H. Vucetich, *Phys. Rev.* **D30** (1984) 363.
21. H. Leutwyler, *Phys. Lett.* **153B** (1985) 65.
22. R. Jackiw and R. Rajaraman, *Phys. Rev. Lett.* **54** (1985) 1219.
23. M. B. Green and J. H. Schwarz, *Phys. Lett.* **149B** (1984) 117.